

significant scientist there. When Kelly wrote his paper in 1956, Shannon refereed it. When I told Shannon about my blackjack betting system, he told me to look at Kelly's paper in deciding how much to bet because in favorable situations, you will want to bet more than in unfavorable situations. I read the Kelly paper, and it made a lot of sense to me.

The Kelly criterion of what fraction of your capital to bet seemed like the best strategy over the long run. When I say long run, a week playing blackjack in Vegas might not sound very long. But long run refers to the number of bets that are placed, and I would be placing thousands of bets in a week. I would get to the long run pretty fast in a casino. In the stock market, it's not the same thing. A year of placing trades in the stock market will not be a long run. But there are situations in the stock market where you get to the long run pretty fast—for example, statistical arbitrage. In statistical arbitrage, you would place tens or hundreds of thousands of trades in a year. The Kelly criterion is the bet size that will produce the greatest expected growth rate in the long term. If you can calculate the probability of winning on each bet or trade and the ratio of the average win to average loss, then the Kelly criterion will give you the optimal fraction to bet so that your long-term growth rate is maximized.

The Kelly criterion will give you a long-term growth trend. The percentage deviations around that trend will decline as the number of bets increases. It's like the law of large numbers. For example, if you flip a coin 10 times, the deviation from the expected value of five will by definition be small—it can't be more than five—but in percentage terms, the deviations can be huge. If you flip a coin 1 million times, the deviation in absolute terms will be much larger, but in percentage terms, it will be very small. The same thing happens with the Kelly criterion: in percentage terms, the results tend to converge on the long-term growth trend. If you use any other criterion to determine bet size, the long-term growth rate will be smaller than for the Kelly criterion. For betting in casinos, I chose the Kelly criterion because I wanted the highest long-term growth rate. There are, however, safer paths that have smaller drawdowns and a lower probability of ruin.

I understand that if you know your edge and it is precisely defined—which of course is not true in the markets—then the Kelly criterion is